

Packet Tracer

IoT Devices

Config and Programming

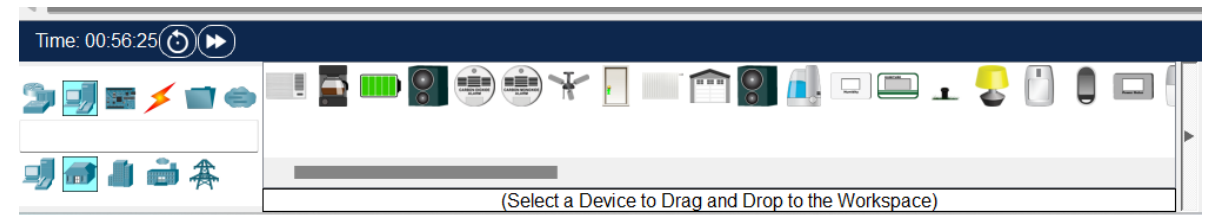


Agenda

- IoT Devices in Packet Tracer
- Programming IoT Devices in PT
- Linking PT simulation to real world!

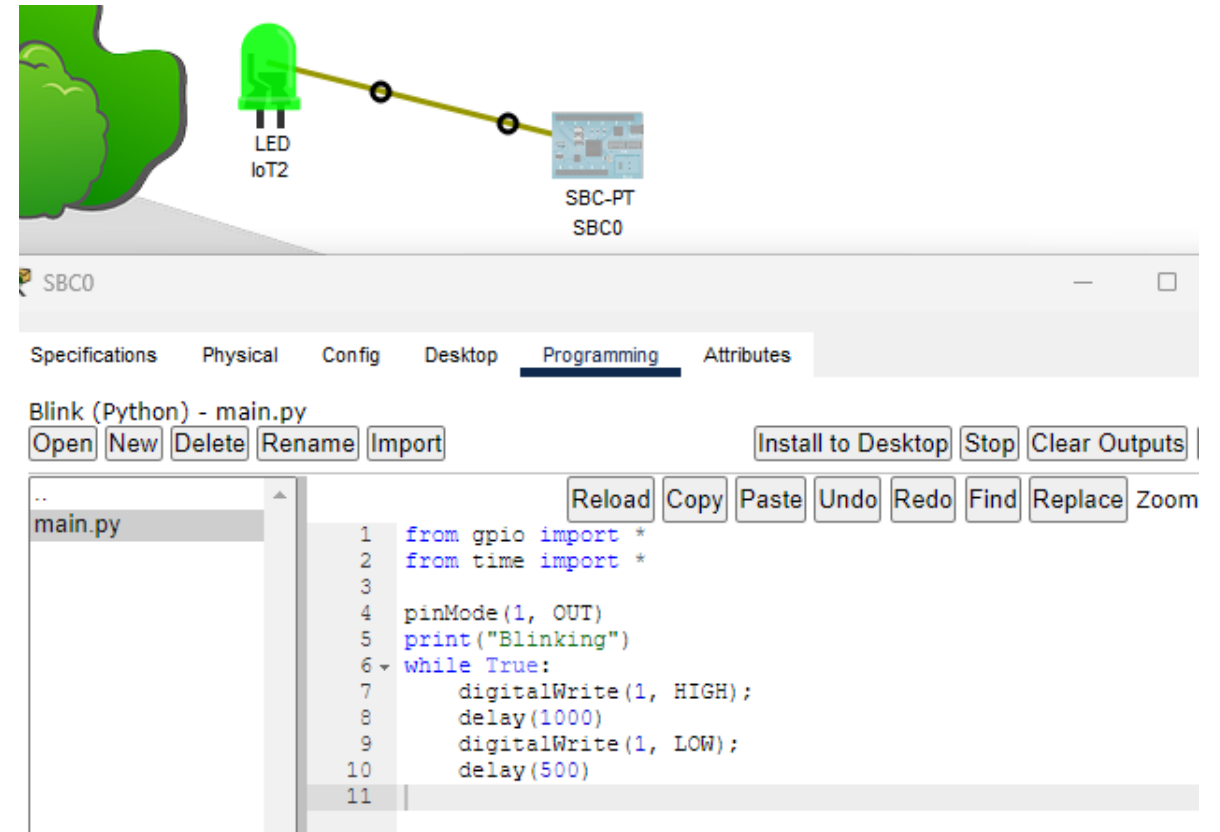
IoT Components in Packet Tracer

- Packet Tracer can be used as a prototyping tool for IoT.
- Packet Tracer can simulate smart networks that make use of IoT devices.
- Many IoT devices for a Smart Home/Industrial/Power Grid network.
- Can build your own “custom” IoT device
- Direct interaction with a device:
 - Hold down the Alt key and click on the device to turn it on or off.
 - Control remotely over the network via a web browser



MCU/SBC/Sensor and Actuators

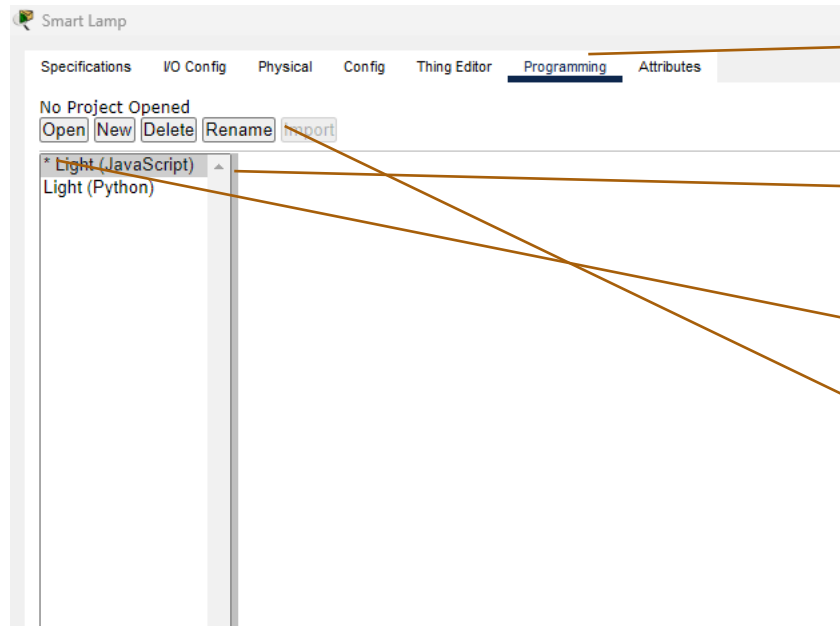
- PT contains a set of common sensors and actuators (components)
 - LED, Button
- Components can't be directly connected to the Network
- Connect to Single Board Computer or Microcontroller, you can create/prototype electronic circuits.
- Connect them using IoT Custom Cable



Program a Device or Custom Component

- Packet Tracer provides support for JavaScript, Python and Visual Programming language called Blockly for most IoT devices/components, including Custom devices
- Access the Programming environment via advanced->Programming tab
- Usually, a default example Project(s) present
- Can create new programming project or edit an existing one

Programming: Projects



1. Select Programming Tab

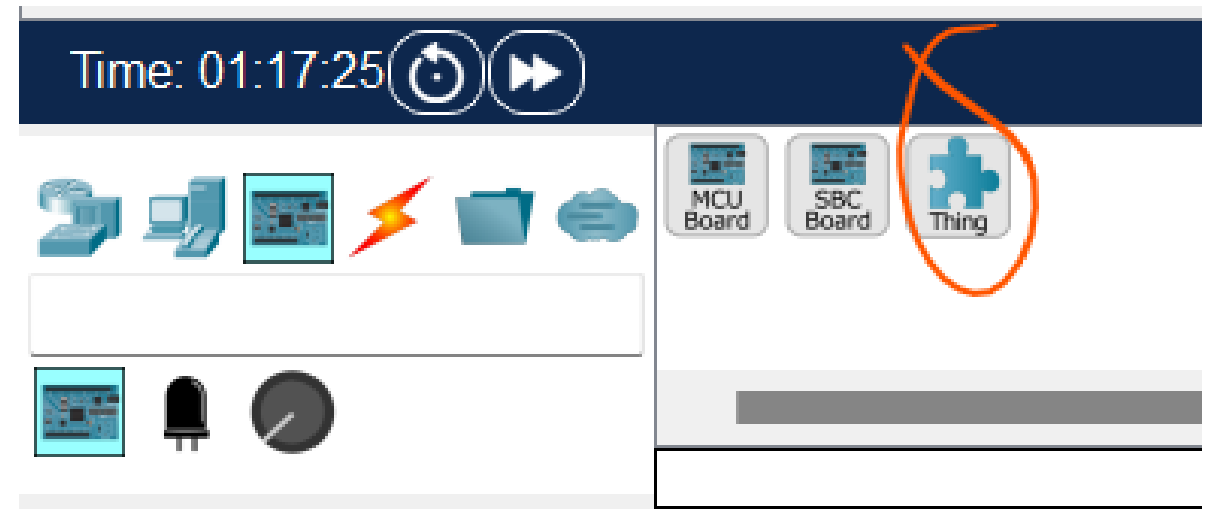
2. Left Panel shows projects available on device

3. Asterix indicates which Project is running

4. Use buttons to Open/Delete/Create/Stop/Start projects

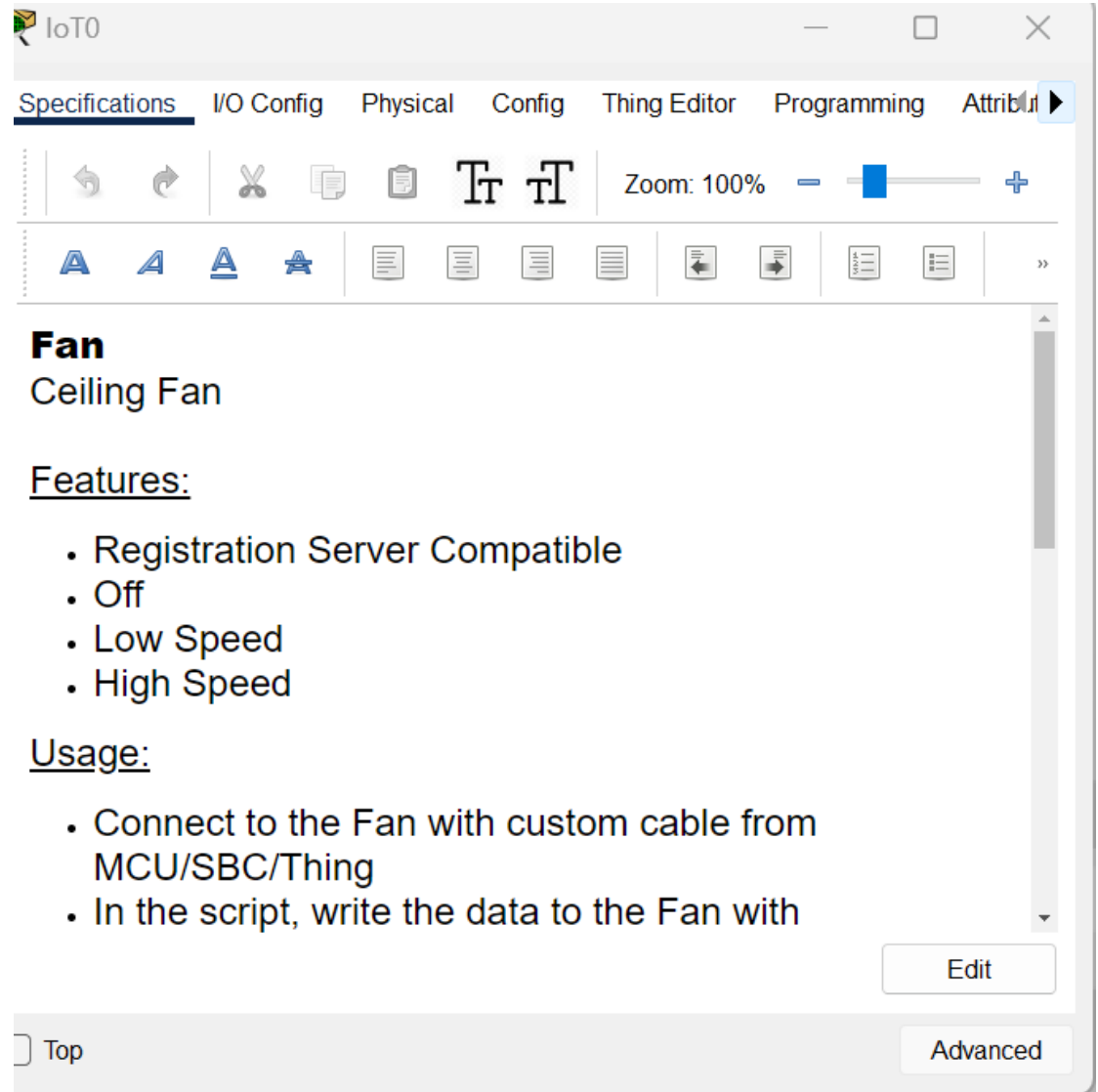
Create a new Component

- To Create a new Thing:
 - Add a Thing from Components list
 - Click the Advanced to access the /Thing Editor
 - Associate the new graphics to their respective states(e.g. on/off)
 - Click on the Config tab to select the network adaptor to be used to connect to the network (if applicable).
- Save it for later:
 - Select Tools/Custom Device Dialog from the toolbar.
 - Click on Select and click on the Thing to be saved.
 - Modify the template and description.
 - Click on the type of new Thing.
 - Click on Add - The new template will be saved in the PT template file on the local disk and the customized Thing will now display with the other sensors.



IoT Components: Configuration

- The Config window of a device allows you to:
 - Specifications - describes the features, usage, local and remote control of the device.
 - Physical - available modules and power connections.
 - Config - show display name, serial number, network configuration and IoT server.
 - Attributes - display the devices attributes such as MTBF, power consumption and cost.
- Advanced config also possible including programming/behaviour



Packet Tracer Environment Controls

- The Physical Workspace uses containers.
- Each container (intercity, buildings, and wiring closets) can have their own set of environmental values.
- There are 24 default environmental elements, such as temperature, rain, water level, wind speed, and snow.
 - A heater will increase ambient temperature when turned on.
 - A smoke detector can be used to trigger an alarm when the smoke in environment increases to a certain point.

The screenshot displays the 'Environments' control panel in Packet Tracer. At the top, the 'Location' is set to 'Intercity' with an 'Edit' button. Below this, the 'Current Time' is '03:00:00', also with 'Edit' and 'Pause' buttons. A section titled 'Select an environment to show its chart.' includes a 'Filter...' input field and 'Search' and 'Reset' buttons. The main area lists 24 environmental parameters, grouped into several categories with expandable/collapsible icons (▴/▾):

- Earth Physical Features**
 - Elevation: 22.00 m
 - Soil pH: 7.00 pH
- Gases**
 - Argon: 0.9340 %
 - CO: 0.00 %
 - CO₂: 0.0360 %
 - He: 0.0005240 %
 - H: 0.00050 %
 - Methane: 0.000150 %
 - Nitrogen: 78.0840 %
 - O₂: 20.9460 %
- Other**
 - Atmospheric Pressure: 101.3250 kPa
- Temperature**
 - Ambient Temperature: 1.00 C
- Water**
 - Clouds: 6.25 %
 - Humidity: 77.50 %
 - Rain: 0.25 cm
- Gravity**
 - Gravity: 9.80 m
- Light (Sun)**
 - Electromagnetic Radiation: 0.00 %
 - Infrared: 0.00 %
 - Radiant Heat: 0.00 %
 - Sunlight: 0.00 %
 - Ultraviolet: 0.00 %
 - Visible: 0.00 %
- Wind**
 - Direction: 26.25 d
 - Variance (gusts): 5.00 %
 - Speed: 0.00 kph

Programming: Creating a new Project

- Select the Template to suit
 - Many available in Python/Javascript
- Modify the code to suit your means
 - Good idea to look at similar devices code as an example

